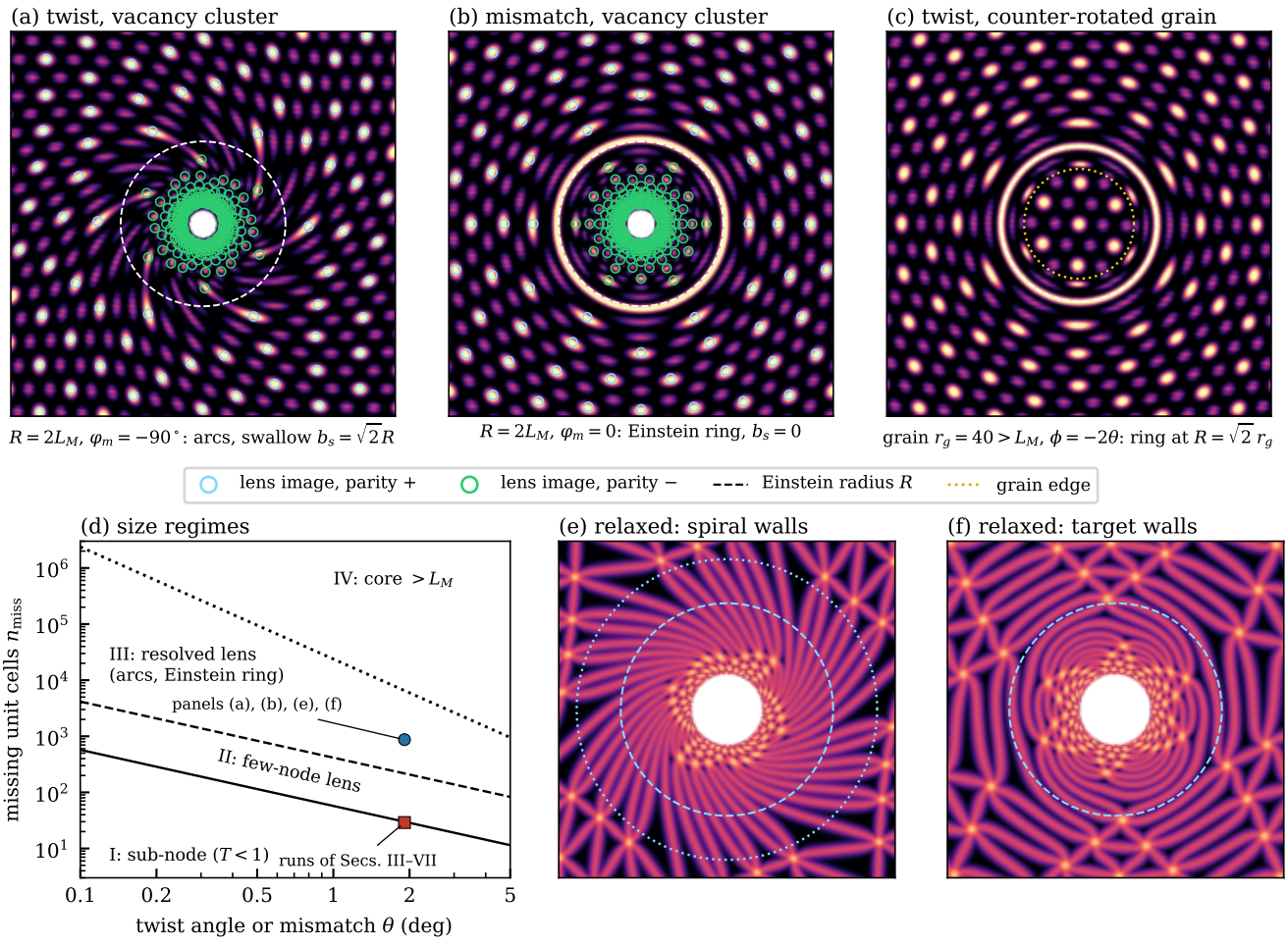

(a) twist, vacancy cluster
(b) mismatch, vacancy cluster
(c) twist, counter-rotated grain
$R = 2L_M$, $\varphi_m = -90^\circ$: arcs, swallow $b_s = \sqrt{2}R$
$R = 2L_M$, $\varphi_m = 0$: Einstein ring, $b_s = 0$
grain $r_g = 40 > L_M$, $\phi = -2\theta$: ring at $R = \sqrt{2}\, r_g$
lens image, parity +
lens image, parity −
Einstein radius R
grain edge
(d) size regimes
missing unit cells n_{miss}
10^6
10^5
10^4
10^3
10^2
10^1
IV: core $> L_M$
III: resolved lens
(arcs, Einstein ring)
panels (a), (b), (e), (f)
II: few-node lens
I: sub-node ($T < 1$)
runs of Secs. III–VII
0.1
0.2
0.5
1
2
5
twist angle or mismatch θ (deg)
(e) relaxed: spiral walls
(f) relaxed: target walls